DRESS: A Continuous Framework for Structural Graph Refinement

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Abstract

The Weisfeiler-Lehman (WL) hierarchy is a cornerstone framework for graph isomorphism testing and structural analysis. However, scaling beyond 1-WL to 3-WL and higher requires tensor-based operations that scale as $\mathcal{O}(n^3)$ or $\mathcal{O}(n^4)$, making them computationally prohibitive for large graphs. In this paper, we start from the Original-DRESS equation (Castrillo, León, and Gómez, 2018)—a parameter-free, continuous dynamical system on edges—and show that it distinguishes the prism graph from $K_{3,3}$, a pair that 1-WL provably cannot separate. We then generalize it to Motif-DRESS, which replaces triangle neighborhoods with arbitrary structural motifs and converges to a unique fixed point under three sufficient conditions, and further to Generalized-DRESS, an abstract template parameterized by the choice of neighborhood operator, aggregation function and norm. Finally, we introduce ∇-DRESS, which *marks* each vertex by applying a multiplicative weight to its incident edges, connecting the framework directly to the WL hierarchy, and Δ-DRESS, which runs DRESS on each vertex-deleted subgraph $G \setminus \{v\}$, connecting the framework to the Kelly–Ulam reconstruction conjecture. Both ∇-DRESS and Δ-DRESS empirically distinguish Strongly Regular Graphs (SRGs)—such as the Rook and Shrikhande graphs—that confound 3-WL. Our results establish the DRESS family as a highly scalable framework that empirically surpasses both 1-WL and 3-WL on well-known benchmark graphs, without the prohibitive $\mathcal{O}(n^4)$ computational cost.

1 Introduction

The graph isomorphism problem and structural similarity measurement rely heavily on the Weisfeiler-Lehman (WL) hierarchy [11]. While the standard 1-WL test—shown to characterize the expressiveness of message-passing neural networks [12, 9]—is highly efficient, it fails to distinguish many non-isomorphic graphs. Higher-order k-WL tests [2] offer greater expressiveness, but their $\mathcal{O}(n^3)$ or $\mathcal{O}(n^4)$ complexity renders them impractical for large-scale network analysis.

The Original-DRESS algorithm [3] provides a continuous alternative: a parameter-free, non-linear dynamical system that iteratively refines edge values based on common-neighbor aggregation. As we show in Section 3, Original-DRESS distinguishes graphs that 1-WL cannot. However, it still fails to distinguish highly symmetric structures such as Strongly Regular Graphs (SRGs).

In this work, we present a hierarchy of continuous structural refinement algorithms, building from the concrete to the general:

1. **Original-DRESS** (Section 2): The foundational equation, a parameter-free dynamical system on edges.

2. **Original-DRESS vs. 1-WL** (Section 3): We show that Original-DRESS distinguishes graphs that 1-WL cannot, including the prism graph from $K_{3,3}$.

3. **Motif-DRESS** (Section 4): A generalization that replaces triangle neighborhoods with arbitrary structural motifs. Original-DRESS is recovered as a special case with $M = K_3$.

4. **Generalized-DRESS** (Section 5): The most abstract template, opening the aggregation function and norm as additional free parameters.
5. **∇-DRESS** (Section 6): Marks each vertex by applying a multiplicative weight to its incident edges, connecting the framework directly to the WL hierarchy.
6. **Δ-DRESS** (Section 7): Runs DRESS on each vertex-deleted subgraph $G \backslash \{v\}$, connecting the framework to the reconstruction conjecture.

Isomorphism Test. Throughout this paper, the **graph fingerprint** is the sorted vector of converged edge values $\mathrm{sort}(d^*)$. Two graphs are declared non-isomorphic if and only if their fingerprints differ element-wise beyond a numerical tolerance ϵ. This applies uniformly to all DRESS variants. An equivalent representation is the **histogram** $h(d^*)$: since all DRESS values lie in $[0, 2]$ (see the convergence proof in Section 4.3) and the convergence tolerance is $\epsilon = 10^{-6}$, each value maps to one of $\lceil 2/\epsilon \rceil = 2 \times 10^6$ integer bins, yielding a fixed-size bin-count vector that uniquely identifies the same multiset. In general, the number of bins is determined by the convergence tolerance: a tolerance of ϵ over the range $[0, 2]$ gives $\lceil 2/\epsilon \rceil$ bins. The histogram is particularly useful when fingerprints from multiple subgraphs must be pooled (see ∇-DRESS and Δ-DRESS, Sections 6–7).

All experiments use a convergence tolerance of $\epsilon = 10^{-6}$ and a maximum of 20 iterations; in every case convergence was reached before the iteration limit.

Related Work. Higher-order GNN architectures that exceed 1-WL expressiveness include k-GNN [9], GIN [12], PPGN [8], and Ring-GNN [5]. Subgraph-based approaches such as GNN-AK+ [13] and ESAN [1] achieve higher expressiveness by running message-passing networks on vertex-deleted or vertex-marked subgraphs; Δ-DRESS (Section 7) shares the vertex-deletion strategy but replaces learned message passing with deterministic fixed-point iteration. All of the above are **supervised** methods that require labeled training data and learnable parameters. In contrast, DRESS is a deterministic, unsupervised framework that produces canonical fingerprints via fixed-point iteration, without any learning or parameter tuning.

2 Original-DRESS

The Original-DRESS equation [3] is a parameter-free, non-linear dynamical system that iteratively refines edge values based on common-neighbor aggregation.

$$d_{uv}^{(t+1)} = \frac{\sum_{x \in N[u] \cap N[v]} \left(d_{ux}^{(t)} + d_{xv}^{(t)} \right)}{\|u\|^{(t)} \cdot \|v\|^{(t)}} \tag{1}$$

where $N[u] = N(u) \cup \{u\}$ denotes the closed neighborhood of u, and $\|u\| = \sqrt{\sum_{x \in N[u]} d_{ux}^{(t)}}$ is the vertex norm. The self-similarity $d_{uu} = 2$ is invariant under iteration. This equation converges to a unique fixed point, providing a continuous structural fingerprint for the graph.

Self-loops. Self-loops are added to every vertex before iteration (i.e., the algorithm uses the closed neighborhood $N[u] = N(u) \cup \{u\}$). The self-loop edge (u, u) participates in both the aggregation and the vertex norm; without it, an isolated edge with no common neighbors would produce $\|u\| \cdot \|v\| = 0$, making the iteration undefined.

Initialization. The equation is scale-invariant (degree-0 homogeneous), so the fixed point is independent of the initial scale. Any uniform initialization $d_{uv}^{(0)} = c$ for all edges, with $c > 0$, converges to the same unique fixed point d^*. In practice, $c = 1$ is used.

However, because Original-DRESS aggregates strictly over triangles (common neighbors), it cannot distinguish graphs with identical triangle counts per edge (e.g., Strongly Regular Graphs).

3 Original-DRESS Distinguishes Graphs That 1-WL Cannot

We now demonstrate a key expressiveness result: Original-DRESS—the simplest member of the DRESS family—already distinguishes graphs that 1-WL (color refinement) cannot.

Theorem 1 (DRESS distinguishes beyond 1-WL)**.** *There exist graph pairs that 1-WL cannot distinguish but Original-DRESS can. In particular, DRESS distinguishes the prism graph ($C_3 \square K_2$) from the complete bipartite graph $K_{3,3}$, a pair that 1-WL provably cannot separate.*

Proof. Both graphs are 3-regular on 6 vertices with 9 edges. Since all vertices have the same degree, 1-WL assigns a uniform color to every vertex in both graphs after every iteration, and therefore cannot distinguish them [2].

DRESS operates on *edges* and aggregates over common neighbors (triangles). We show that the prism must have at least two distinct edge values at convergence, while $K_{3,3}$ has a single uniform value.

$K_{3,3}$: No two adjacent vertices share a common neighbor ($K_{3,3}$ is triangle-free). Hence every edge has the same neighborhood structure, and by symmetry the unique fixed point assigns the same value to all 9 edges. Numerically, $d^*_{K_{3,3}} \approx 1.155$ for all edges.

Prism graph: There are two structurally distinct edge types: *triangle edges* (6 edges forming the two triangular faces; each pair of endpoints shares 1 common neighbor) and *matching edges* (3 edges connecting corresponding vertices of the two triangles; each pair of endpoints shares 0 common neighbors besides self-loops). Suppose for contradiction that all 9 edges converge to the same value d^*. Then:

- Triangle edges: $d^* = \dfrac{4 + 4d^*}{2 + 3d^*}$, which gives $d^* = \frac{1+\sqrt{13}}{3} \approx 1.535$.
- Matching edges: $d^* = \dfrac{4 + 2d^*}{2 + 3d^*}$, which gives $d^* = \frac{2}{\sqrt{3}} \approx 1.155$.

These are contradictory, so the prism cannot have a uniform fixed point. The converged unique edge values are:

Graph	**Unique values**	**Multiplicities**
Prism	$\{0.922,\ 1.709\}$	3 matching edges, 6 triangle edges
$K_{3,3}$	$\{2/\sqrt{3} \approx 1.155\}$	all 9 edges identical

Since 1-WL cannot distinguish them and DRESS can, DRESS distinguishes beyond 1-WL on this instance. □

Remark. This result holds for Original-DRESS alone—the simplest member of the DRESS family, using only triangle neighborhoods and $\mathcal{O}(E)$ computation per iteration. The generalizations that follow (Motif-DRESS, ∇-DRESS, Δ-DRESS) extend this advantage further to graphs that resist even 3-WL.

4 Motif-DRESS

Original-DRESS is limited to triangle neighborhoods. What if we use other structural motifs? Motif-DRESS generalizes Original-DRESS by defining $\mathcal{N}(u, v)$ as the set of edges that co-occur with (u, v) in a specific structural motif M, with $f = \text{sum}$, $g =$ geometric norm, and an optional symmetric weight function $w : E \to \mathbb{R}_{>0}$:

$$d_{uv}^{(t+1)} = \frac{\sum_{e' \in \mathcal{N}_M(u,v)} w_{e'}\, d_{e'}^{(t)}}{\|u\|^{(t)} \cdot \|v\|^{(t)}} \tag{2}$$

where $\|u\|^{(t)} = \sqrt{\sum_{e' \in \mathcal{N}_M(u,u)} w_{e'}\, d_{e'}^{(t)}}$ is the weighted vertex norm at time t.

The symmetric weight function $w(e)$ acts as a multiplicative factor, controlling how much structural information flows along each edge. Because the weights appear identically in the numerator and denominator (both are degree-1 in $w \cdot d$), uniformly scaling all weights does not change the fixed point; only the relative weights matter.

Different choices of motif M yield different neighborhood operators:

- **Triangle ($M = K_3$):** $\mathcal{N}(u,v) = \{(u,x),(x,v) \mid x \in N[u] \cap N[v]\}$—this recovers Original-DRESS.
- **4-cycle ($M = C_4$):** Aggregates over edges participating in 4-cycles with (u,v).
- **K_4 clique:** Aggregates over edges forming a K_4 with (u,v).

4.1 Properties

Complexity: $\mathcal{O}(\text{Motif Extraction}) + \mathcal{O}\big(I \cdot \sum_e |\mathcal{N}_M(e)|\big)$, where I is the number of iterations and $\sum_e |\mathcal{N}_M(e)|$ is the total size of all motif-neighborhood lists. For Original-DRESS (triangles), this reduces to $\mathcal{O}(I \cdot m \cdot T)$, where T is the number of triangles. For motifs such as 4-cycles or K_4 cliques in sparse graphs, the motif-neighborhood lists can be significantly smaller, making Motif-DRESS faster than Original-DRESS.

Invariant: $d_{uu} = 2$. The self-similarity is invariant under iteration: since $\mathcal{N}_M(u,u)$ includes all edges incident to u symmetrically, and $d_{ux} = d_{xu}$, substituting $v = u$ yields $d_{uu}^{(t+1)} = 2\|u\|^2/\|u\|^2 = 2$ for all t. Thus $d_{uu} = 2$ is a fixed property of the equation, not a free parameter.

4.2 Expressiveness

All experiments below use the K_4 clique motif. The specific SRG pairs tested below are known to be indistinguishable by 3-WL; each successful distinction therefore demonstrates that Motif-DRESS empirically exceeds 3-WL on these instances.

- **Rook vs. Shrikhande:** Successfully distinguishes this pair of SRGs with parameters $(16,6,2,2)$. The Rook graph ($K_4 \square K_4$) contains K_4 cliques while the Shrikhande graph does not, so the K_4-neighborhood sizes differ per edge.
- **Chang Graphs:** Distinguishes all 6 pairwise comparisons among the four SRGs with parameters $(28,12,6,4)$: T(8), Chang-1, Chang-2, and Chang-3.

4.3 Proof Sketch of Convergence

Theorem 2 (Convergence of Motif-DRESS)**.** *Let $F : \mathbb{R}_+^{|E|} \to \mathbb{R}_+^{|E|}$ be the Motif-DRESS update operator. If the aggregation function and the vertex norm satisfy the degree-0 homogeneity condition ($p = 2q$), then F converges to a unique fixed point $d^* \in [0,2]^{|E|}$ for any non-negative initial state $d^{(0)} > 0$.*

Proof sketch. The argument proceeds in three steps:

1. **Scale Invariance (Degree-0 Homogeneity):** The aggregation function is positively homogeneous of degree p, and the norm of degree q. Since $p = 2q$, the ratio $F(d)$ is homogeneous of degree 0: for any $\lambda > 0$, $F(\lambda d) = F(d)$. Hence the iteration is self-regularizing and cannot diverge or collapse to zero.
2. **Boundedness:** For any valid symmetric neighborhood operator $\mathcal{N}_M(e)$, the Cauchy–Schwarz inequality (or Hölder's inequality for Minkowski-r variants) bounds the numerator by the denominator. Self-loop inclusion ensures the denominator is strictly positive. Thus $F(d)_{uv} \in [0,2]$ for all $d > 0$.

3. **Contraction on the Hilbert Projective Metric:** Since F is a positive, degree-0 homogeneous map on the cone $\mathbb{R}^{|E|}_{>0}$, Birkhoff's contraction theorem guarantees that F is a strict contraction under the Hilbert projective metric $d_H(x, y) = \log\big(\max_e x_e/y_e \cdot \max_e y_e/x_e\big)$, provided F maps a bounded part of the cone into a strictly smaller part—which follows from the Cauchy–Schwarz bound above. By the Banach fixed-point theorem on the complete metric space $(\mathbb{R}^{|E|}_{>0}/\sim, d_H)$, the iteration converges to a unique ray, and the boundedness step pins the representative to $d^* \in [0, 2]^{|E|}$.

A complete formal verification of the contraction constant is deferred to future work; all empirical tests in this paper confirm convergence within 20 iterations. □

5 Generalized-DRESS

Motif-DRESS fixes the aggregation to summation and the norm to the geometric mean. Generalized-DRESS is the most abstract template, allowing any choice of these components. For each edge (u, v) and a symmetric neighborhood operator $\mathcal{N}(u, v)$:

$$d_{uv}^{(t+1)} = \frac{f\big(\{d_{e'}^{(t)}\}_{e' \in \mathcal{N}(u,v)}\big)}{g^{(t)}(u) \cdot g^{(t)}(v)} \tag{3}$$

where:

- $\mathcal{N}(u, v)$ is a **symmetric neighborhood operator**—the set of edges aggregated for edge (u, v),
- f is the **aggregation function** over edges in $\mathcal{N}(u, v)$,
- $g(u)$ is the **vertex norm**.

Original-DRESS and Motif-DRESS are both special cases: Original-DRESS fixes $\mathcal{N}$ to triangles, f = sum, g = geometric norm; Motif-DRESS generalizes $\mathcal{N}$ to arbitrary motifs while keeping the same f and g. Generalized-DRESS opens all three parameters, enabling variants such as Cosine-DRESS (cosine similarity aggregation) or Minkowski-r norms.

6 ∇-DRESS: Individualization

∇-DRESS breaks symmetry by *marking* a vertex: applying a distinguished multiplicative weight $\alpha \neq 1$ on its incident edges, while preserving the full graph topology.

Formally, given a graph $G = (V, E, w)$, a DRESS variant $\mathcal{F}$, and an individualization weight $\alpha \neq 1$, the individualized graph $G^{(\alpha,v)}$ is obtained by multiplying the weight of every edge incident to v by α:

$$w^{(\alpha,v)}(e) = \begin{cases} \alpha \cdot w(e) & \text{if } v \in e, \\ w(e) & \text{otherwise.} \end{cases}$$

The ∇-DRESS fingerprint is then the multiset of per-vertex DRESS fingerprints:

$$\nabla\text{-DRESS}(\mathcal{F}, G, \alpha) = \{\{\text{sort}(\mathcal{F}(G^{(\alpha,v)})) : v \in V\}\}$$

Equivalently, a pooled histogram can be formed by accumulating all converged edge values across all n individualizations.

Individualizing a vertex in a regular graph breaks the uniform degree structure—marked edges receive a distinct weight, causing the DRESS iteration to resolve structural differences that were invisible under uniform weights.

6.1 Connection to the WL Hierarchy

The key theoretical advantage of ∇-DRESS is its direct connection to the WL hierarchy: $(j+1)$-WL is equivalent to running j-WL over all single-vertex individualizations [2, 6]. The companion paper [4] develops this connection formally.

6.2 Properties

Complexity: $\mathcal{O}(n \cdot I \cdot m \cdot d_{\max})$. For each vertex, ∇-DRESS patches only the incident edge weights (an $\mathcal{O}(d_{\max})$ operation), then runs a standard DRESS iteration on the full graph. The n individualized runs are entirely independent, making ∇-DRESS embarrassingly parallel.

Weight parameter: The individualization weight α can be any value $\neq 1$; the default is $\alpha = 2$. Because DRESS is a contraction on the Hilbert projective metric, the fixed point is a continuous function of the weights, so the specific value of α affects the numerical fingerprint but not the qualitative distinguishing ability (any $\alpha \neq 1$ breaks the same symmetries).

6.3 Expressiveness

∇-DRESS empirically distinguishes the following non-isomorphic pairs:

- **Rook vs. Shrikhande:** Successfully distinguished (SRG$(16, 6, 2, 2)$; confounds 3-WL).
- **Chang Graphs:** Distinguished all 6 pairs among T(8) and the three Chang graphs (SRG$(28, 12, 6, 4)$; confound 3-WL).
- $2 \times C_4$ **vs.** C_8**:** Successfully distinguished (both 2-regular on 8 vertices).
- **Petersen vs. Pentagonal Prism:** Successfully distinguished (both 3-regular on 10 vertices).

7 Δ-DRESS: Deletion

Δ-DRESS breaks symmetry by running DRESS on each vertex-deleted subgraph $G \setminus \{v\}$ for every $v \in V$. The Δ-DRESS fingerprint is the multiset of per-vertex DRESS fingerprints, or equivalently a pooled histogram accumulating all converged edge values across all n deletions.

Deleting a vertex from a regular graph produces an irregular subgraph where DRESS can now distinguish structure that was hidden by the uniform regularity.

7.1 Connection to the Reconstruction Conjecture

The multiset $\{\{\text{DRESS}(G \setminus \{v\}) : v \in V\}\}$ is directly analogous to the *deck* in the Kelly–Ulam reconstruction conjecture, which posits that graphs with $n \geq 3$ are determined (up to isomorphism) by their multiset of vertex-deleted subgraphs. Δ-DRESS computes a continuous relaxation of this deck.

7.2 Relationship to ∇-DRESS

Vertex individualization is at least as informative as vertex deletion: from the individualized graph $G^{(\alpha,v)}$, one can identify the marked edges and recover the structure of $G \setminus \{v\}$, but not vice versa. The Reconstruction Conjecture [7, 10] asserts that for $n \geq 3$, the deck of vertex-deleted subgraphs determines the graph up to isomorphism; if true, the two operators are equivalent in discriminative power.

7.3 Properties

Complexity: $\mathcal{O}(n \cdot I \cdot m \cdot T)$, where T is the number of triangles and I is the number of iterations per deletion. The computation for each deleted subgraph is entirely independent, making Δ-DRESS embarrassingly parallel.

7.4 Expressiveness

Δ-DRESS empirically produces identical outcomes to ∇-DRESS on all tested graph pairs, including all SRG pairs listed above.

8 Family Structure

The DRESS variants introduced in this paper form a nested hierarchy with two orthogonal composition operators:

$$\text{Generalized-DRESS} \supset \text{Motif-DRESS} \supset \text{Original-DRESS}$$

$\nabla(\cdot)$ and $\Delta(\cdot)$ are **orthogonal wrappers** applicable to any of the above: given any DRESS variant $\mathcal{F}$,

$$\nabla\text{-DRESS}(\mathcal{F}, \alpha) = \{\{\mathcal{F}(G^{(\alpha,v)}) : v \in V\}\} \qquad \Delta\text{-DRESS}(\mathcal{F}) = \{\{\mathcal{F}(G \setminus \{v\}) : v \in V\}\}$$

Both strategies are independent of the choice of $\mathcal{N}$, f, and g. ∇-DRESS breaks symmetry by weight individualization; Δ-DRESS by deletion. Both generalize to depth k: ∇^k individualizes all $\binom{n}{k}$ vertex subsets, and Δ^k runs DRESS on all $\binom{n}{k}$ vertex-deleted subgraphs. The exploration of this iterated hierarchy and its relationship to the WL hierarchy is developed in the companion paper [4].

9 Conclusion

We presented the DRESS family, a hierarchy of continuous structural refinement algorithms. Starting from Original-DRESS—a single equation proposed in 2018—we showed it distinguishes graphs that 1-WL cannot (Theorem 1). We then generalized it to Motif-DRESS, which replaces triangle neighborhoods with arbitrary structural motifs and converges to a unique fixed point under three sufficient conditions. Generalized-DRESS abstracts further, opening the aggregation function and norm as free parameters. ∇-DRESS extends the framework via vertex individualization (multiplicative weight marking), connecting directly to the WL hierarchy, while Δ-DRESS extends it by running DRESS on each vertex-deleted subgraph, connecting it to the Kelly–Ulam reconstruction conjecture. Both empirically distinguish Strongly Regular Graphs that confound 3-WL, while maintaining scalability well below $\mathcal{O}(n^3)$ for sparse graphs. These results establish the DRESS family as a highly scalable framework that empirically surpasses both 1-WL and 3-WL, without the prohibitive computational cost of higher-order WL tests. An open-source implementation with bindings for C, C++, Python, Rust, Go, Julia, R, MATLAB, and WebAssembly is publicly available at `https://github.com/velicast/dress-graph`.

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